

— PUTNAM PROBLEM

Let $k \geq 1$, let $q = 2^k$, and let $\mathbb{F}_q$ be the field with q elements. For each positive integer n , let N_n denote the number of monic polynomials $f \in \mathbb{F}_q[x]$ of degree n such that

$$\gcd(f(x), f(x+1)) = 1.$$

Find N_n for every $n \geq 1$.